

DEPOINDEX: AI-POWERED DEPOSITION TOPIC INDEX

Presented by:

Saumya Dwivedi

Project: DepoIndex (PROBLEM 3)

Source: Persis Yu Deposition

Date: 13 September 2026

Problem:

- Long deposition transcripts are difficult to navigate manually.
- Attorneys need to quickly locate where specific subjects were discussed.
- A topic label is useful only when its page/line citation is verifiable.

Key Requirements:

- Identify meaningful topics and topic transitions
- Determine topic start/end boundaries
- Preserve exact page & line references
- Handle continuing topics and brief digressions
- Provide source evidence for independent verification

Input: Persis Yu Deposition

Output: Topic Index in JSON + Markdown

Objective:

Build a system that converts a deposition transcript into a chronological, verifiable Topic Index.

SYSTEM ARCHITECTURE

Architecture Principle:

LLM generates topics, but source references remain tied to the original transcript.

Core Design Goals:

- Evidence-grounded topic labels
- Exact source addressability
- Chronological ordering
- Attorney-verifiable output

DEPOINDEX PIPELINE



TOPIC SEGMENTATION & PROVENANCE

Topic Segmentation:

The system must distinguish between:

- New topic → genuine subject-matter change
- Continuation → same subject continues
- Digression → short deviation within a larger topic
- Re-entry → earlier topic returns later
- Related subject → closely connected but independently useful topic

Verification Principle: A plausible topic with the wrong page/line is treated as a failure.

Boundary Strategy:

Candidate boundaries are based on changes in the testimony's subject matter and discourse flow.

The generated topic label should be grounded in the evidence inside its own citation range, rather than unrelated surrounding context.

Provenance Strategy: Each entry stores:

- Topic
- Start Page + Line
- End Page + Line
- Supporting Evidence / Source Reference



EXAMPLE & VALIDATION RESULTS



Example Topic Index Entry

Topic	Start	End
 Vervent Role & Loan Servicing Operations	p. 24:5	p. 24:25n

Supporting evidence: Original deposition testimony at the cited location.

Overall Assessment:

PROTOTYPE-STAGE:

Strong source-location accuracy, but topic segmentation and evidence-grounded labeling require further refinement.

Main finding: 20/20 sampled page/line locations were correctly mapped to the actual Persis Yu PDF.

Manual Validation

Manual Validation — 20 Topic Index Entries

Criterion	PASS	PARTIAL	FAIL
Location Accuracy	20	0	0
Topic Relevance	15	3	2
Boundary Quality	12	5	3
Coverage	17	2	1
Redundancy	13	4	3

STABILITY, FAILURES & NEXT STEPS

Stability Test:

Observed topic counts:

Run 1 → 20 raw entries

Run 2 → 15 topics

Run 3 → 9 topics

Verdict: UNSTABLE

Important limitation:

Run 2 and Run 3 were manual proxy re-segmentations, not genuine automated re-runs of the pipeline.

Next Steps:

- Run the actual pipeline 3+ times with deterministic settings
- Add automated provenance verification
- Add merge/de-duplication
- Establish fixed segmentation granularity
- Improve evidence-grounded topic labeling

Three Major Failure Cases:

1. Nested Topic Ranges

- Some entries were completely contained within neighboring entries.
- Fix: Detect nested ranges and merge/redraw boundaries where no genuine topic change exists.

2. Incorrect Topic Labels

- Some labels did not match the cited testimony.
- Fix: generate labels strictly from the cited evidence + controlled topic vocabulary.

3. Overlapping Ranges

- Some neighboring entries claimed overlapping testimony.
- Fix: Add overlap detection + merge/de-duplication + clean boundary enforcement.

Final Verdict:

Promising prototype — strong location accuracy, but segmentation and topic-label grounding need improvement before unsupervised professional use.